THE INERTIA STACK AND THE RELATIVE INERTIA STACK ARE ALGEBRAIC AND LOCALLY OF FINITE TYPE OVER THE STACK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The equivalence describing the inertia stack and the relative inertia stack as self fibre products of the diagonal. Invoked at the start of the counted proof (source lines 1239-1246) as a cross-chapter pointer. Shows that the inertia stacks are algebraic stacks.
- The statement that the isomorphism space of two objects is an algebraic space locally of finite type. Invoked inside the counted proof (source lines 1256-1259). Makes the pullback of the inertia a space locally of finite type.
- The description of the relative inertia as a fibre product of isomorphism spaces. Invoked inside the counted proof (source lines 1268-1271) as a cross-chapter pointer. Reduces the relative case to a comparison of two isomorphism spaces.
- The permanence statement for locally of finite type morphisms of algebraic spaces. Invoked near the end of the counted proof (source lines 1274-1278) as a cross-chapter pointer. Gives local finite type of the comparison morphism.

Source reference. Original reference categories-lemma-inertia-fibred-category: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1240–1240. Navigation pointer only: the counted text refers to the Stacks item categories-lemma-inertia-fibred-category, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Source reference. Original reference categories-lemma-relative-inertia-as-fibre-product: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1282–1282. Navigation pointer only: the counted text refers to the Stacks item categories-lemma-relative-inertia-as-fibre-product, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-isom-locally-finite-type: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 278–278. Navigation pointer only: the counted text refers to the Stacks item lemma-isom-locally-finite-type, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-permanence-finite-type: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 307–307. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-permanence-finite-type, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Then the inertia stack $\mathcal{I}_\mathcal{X}$ is an algebraic stack as well. The morphism*

$$\mathcal{I}_\mathcal{X} \longrightarrow \mathcal{X}$$

is representable by algebraic spaces and locally of finite type. More generally, let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Then the relative inertia $\mathcal{I}_{\mathcal{X}/\mathcal{Y}}$ is an algebraic stack and the morphism

$$\mathcal{I}_{\mathcal{X}/\mathcal{Y}} \longrightarrow \mathcal{X}$$

is representable by algebraic spaces and locally of finite type.

Proof. By Categories, Lemma categories-lemma-inertia-fibred-category there are equivalences

$$\mathcal{I}_\mathcal{X} \to \mathcal{X} \times_{\Delta, \mathcal{X} \times_S \mathcal{X}, \Delta} \mathcal{X} \quad \text{and} \quad \mathcal{I}_{\mathcal{X}/\mathcal{Y}} \to \mathcal{X} \times_{\Delta, \mathcal{X} \times_\mathcal{Y} \mathcal{X}, \Delta} \mathcal{X}$$

which shows that the inertia stacks are algebraic stacks. Let $T \to \mathcal{X}$ be a morphism given by the object x of the fibre category of $\mathcal{X}$ over T. Then we get a 2-fibre product square

$$\begin{array}{ccc} Isom_\mathcal{X}(x, x) & \longrightarrow & \mathcal{I}_\mathcal{X} \\ \downarrow & & \downarrow \\ T & \xrightarrow{x} & \mathcal{X} \end{array}$$

This follows immediately from the definition of $\mathcal{I}_\mathcal{X}$. Since $Isom_\mathcal{X}(x, x)$ is always an algebraic space locally of finite type over T (see Lemma lemma-isom-locally-finite-type) we conclude that $\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is representable by algebraic spaces and locally of finite type. Finally, for the relative inertia we get

$$\begin{array}{ccccc} Isom_\mathcal{X}(x, x) & \longleftarrow & K & \longrightarrow & \mathcal{I}_{\mathcal{X}/\mathcal{Y}} \\ \downarrow & & \downarrow & & \downarrow \\ Isom_\mathcal{Y}(f(x), f(x)) & \xleftarrow{e} & T & \xrightarrow{x} & \mathcal{X} \end{array}$$

with both squares 2-fibre products. This follows from Categories, Lemma categories-lemma-relative-inertia-as-fibre-product. The left vertical arrow is a morphism of algebraic spaces locally of finite type over T, and hence is locally of finite type, see Morphisms of Spaces, Lemma spaces-morphisms-lemma-permanence-finite-type. Thus K is an algebraic space and $K \to T$ is locally of finite type. This proves the assertion on the relative inertia. $\square$